The Erdős–Straus count is governed by a quadratic L-value

An experimental route from solution-counting to $L(1,\chi_p)$, with a machine-verified account of the wall

J. Naude, with derivations, engines, and analyses produced in human–AI collaboration (Claude, Anthropic; see the repository commit trailers). Computational artifacts, datasets, and machine proofs accompany this paper in `REPORT.md`, `analysis/`, `engines/`, and the Lean 4 development `erdos1/subsetsums/`.

*Working paper, revision of 2026-06-21. The Erdős–Straus conjecture is **open**; nothing here proves or disproves it. The contribution is a measured, reproducible, structurally-identified law for the solution count, a machine-verified delimitation of exactly why the elementary toolbox cannot close the problem, and a complete map of the analytic wall the law runs into.*

Abstract

Let $f(p)$ denote the number of unordered positive solutions of $\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$. Building on the divisor-sum framework of Elsholtz and Tao (2013), we report an empirical law, verified across 278,570 primes of the six hard residue classes mod 840 from 73 to 2.01×10^9 and reproduced here by direct re-execution:

$$f(p) \;\approx\; (\log p)^3 \cdot L(1,\chi_p)^{-c}, \qquad \chi_p = \left(\frac{\cdot}{p}\right), \quad c > 0,$$

where χ_p is the real quadratic character of $\mathbb{Q}(\sqrt{p})$ and $L(1,\chi_p) = 2h(p)\log \varepsilon_p/\sqrt{p}$ is its Dirichlet L-value — the class number times the regulator. Concretely, $\operatorname{corr}\big(\ln f,\ \ln L(1,\chi_p)\big) = -0.62$ ($n = 14{,}955$ primes near 10^9, re-run for this paper; 95% CI $\approx \pm 0.01$, and unchanged under partial correlation controlling for $\log p$), stable across Euler-product truncations $X = 50 \dots 1500$. The exponent c is **not** a fixed constant: measured in narrow windows it *declines monotonically* with scale — from ≈ 0.93 at 10^6 to ≈ 0.46 at 10^{10} (§8.2) — while the correlation itself stays near -0.62, so the scale-robust statement is the **sign** of the coupling, not a numerical exponent. The **more** primes split in $\mathbb{Q}(\sqrt{p})$, the **fewer** Erdős–Straus solutions p has.

We arrive at the law by a documented chain: (i) a cross-validated solution-count dataset and a lognormal law for $f(p)$ confirmed by blind prediction; (ii) a machine-verified kernel reducing $f(p)$ to divisors of shifted integers in residue classes, together with the square obstruction re-proved in full and formalized in Lean 4 + Mathlib (20 theorems, complete with no unproved steps); (iii) a residual spectrum exposing a quadratic-residue ladder $s_q \approx 18\, q^{-1.95}$ — "non-residues richer at every modulus" — which is the first-order shadow of the character χ_p; (iv) a Fourier decomposition of the local divisor density over Dirichlet characters

mod ℓ that isolates the quadratic character (p/ℓ) as the dominant non-trivial signal at every $\ell \in \{11, 13, 17, 19, 23\}$ (14–47σ), whose Euler product is by definition a power of $L(1, \chi_p)$.

The mechanism is the McKee–Zhou identity, by which the singular series of $\sum_{n \leq N} \tau(F(n))$ for an irreducible quadratic F equals $2L(1, \chi_{\mathrm{disc}\, F})/\zeta(2)$ — the precedent that divisor/representation counts are governed by quadratic L-values (Gauss–Siegel). The law is **not** a theorem: deriving the singular series via McKee–Zhou blocks at a verified obstruction (the Type II side-condition $\delta \mid y'z'$ makes f_{II} a *two-shift* divisor correlation, not a single $\sum \tau(F)$), which is the two-shift Titchmarsh estimate flagged out of reach by Elsholtz–Tao and blocked by the classical (Selberg) $\sqrt{x}$ parity barrier. What the L-lens does buy is sharp and unconditional in spirit: any Erdős–Straus exception must lie in the **extreme class-number tail** (a Granville–Soundararajan super-rare set), Siegel zeros are the **best** case rather than the worst, and the size budget — even at the maximal order $L(1, \chi_p) \asymp e^{\gamma} \log\log p$ — never threatens $f > 0$.

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1. Introduction

1.1 The conjecture

The Erdős–Straus conjecture (Erdős and Straus, 1948) asserts that for every integer $n \geq 2$ there exist positive integers x, y, z with

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}.$$

It suffices to prove it for primes p, and solutions are known to exist for every p except possibly those in the six "square classes"

$$p \equiv 1^2,\ 11^2,\ 13^2,\ 17^2,\ 19^2,\ 23^2 \pmod{840},$$

where every covering-congruence identity provably dies (Mordell; Schinzel; Yamamoto). The statement has been verified computationally to 10^{17} (Salez, 2014) and to 10^{18} (Mihnea–Bogdan, 2025); it remains open. Throughout, $p \geq 5$ is prime, and on the hard classes $p \equiv 1 \pmod 4$, so $\mathbb{Q}(\sqrt{p})$ is real with discriminant p and $\chi_p = (\frac{\cdot}{p})$ satisfies $(p/\ell) = (\ell/p)$ by reciprocity.

1.2 What was known

The deepest count-theoretic results are due to **Elsholtz and Tao** (*J. Aust. Math. Soc.* **94** (2013), 50–105; arXiv:1107.1010). Writing $f(p)$ for the number of solutions with $x \leq y \leq z$, they prove, among much else:

- a first-moment bound $N \log^2 N \ll \sum_{p \leq N} f(p) \ll N \log^2 N \log\log N$ (so the *average* order of $f(p)$ is $\log^3 p$, up to a $\log\log$ factor that they note but cannot remove);
- pointwise bounds $f(p) \ll p^{3/5+o(1)}$ and a lower bound $f(p) \geq (\log p)^{0.549}$ valid on a **density-1** set of primes (their Thm 1.8);
- a Type I / Type II decomposition realizing $f(p)$ as a sum of divisor counts of shifted integers;
- the remark (their Remark 1.3) that even the **second moment** $\sum_p f(p)^2$ "seems to ... lie out of reach of our methods, as the level of the relevant divisor sums becomes too great to handle," and a Poisson/Borel–Cantelli heuristic for solubility (Remark 1.2).

The strongest "almost all" statement is **Vaughan (1970)**: the number of $n \leq N$ for which $4/n$ is *not* so expressible is $\ll N \exp\big(-c(\log N)^{2/3}\big)$. The square classes are immune to covering systems by **Schinzel** and **Yamamoto** ($f_I(c^2) = f_{II}(c^2) = 0$ for odd squares, by quadratic reciprocity), so that the conjecture's content is precisely the gap "density $1 \to$ all primes." The absence of a Brauer–Manin obstruction *to the existence of solutions* (Bright–Loughran, 2020 — their Brauer group is a nontrivial $\mathbb{Z}/2\mathbb{Z}$, but it obstructs only strong approximation, not existence) confirms that the difficulty is **analytic, not cohomological**.

1.3 Contribution

This paper's central result is the ***L*-function law** of the abstract: $f(p)$ is modulated by $L(1, \chi_p)$, equivalently by the class number/regulator of $\mathbb{Q}(\sqrt{p})$. To our knowledge this connection is unrecorded — Elsholtz–Tao build $f(p)$ out of exactly the divisor sums whose averages McKee's theorem evaluates, but the count-theoretic literature uses only their order of magnitude, never extracting the constant where (as §8 shows) $L(1, \chi_p)$ lives. (We attribute the singular-series constant to McKee, not to Elsholtz–Tao, who do not invoke it.) The paper's secondary contributions, each a rung on the ladder to the law and each separately machine-checked, are:

1. the largest per-prime solution-count dataset we are aware of, four-engine cross-validated and validated against the published external dataset with zero discrepancies (§3);
2. a lognormal law for $f(p)$, confirmed by four blind predictions of window minima before the computations were run (§3.3);
3. the kernel bijection and the square obstruction, re-proved in full and **formalized in Lean 4 + Mathlib, complete with no unproved steps** (20 theorems; §4, Appendix A), together with a no-finite-channel-system theorem (Theorem F) and an ε-covering cost law;
4. the explicit reduction of the second moment of f_{II} to a two-shift Titchmarsh divisor sum, measured 98.5% off-diagonal, and the one rigorous distributional law in the circle, an Erdős–Kac theorem for the channel

count $\omega_3(4p+1)$ (§6);

5. a heuristic but data-matched derivation of the growth exponent $f(p) \sim (\log p)^3$ from the arithmetic geometry of the Cayley cubic (§7).

We are scrupulous about epistemic status. The L-function law is a **verified, structurally-identified empirical law**, not a theorem; §9 maps in full why a proof meets the same two-shift parity wall as every other route, and which single inequality would change that.

1.4 Method and standards of evidence

The project's working rule is: **no claim without a machine check, no engine without independent validation, no model without a blind test.** Four solution-counting engines were written independently in C, Rust, CUDA, and C/OpenMP; their outputs are byte-identical at five scales and reproduce the external dataset of the 2025 verification preprint on thousands of overlapping primes. Every numbered lemma in the structural sections is checked in exact arithmetic (`analysis/verify_lemmas.py`, 8,719 assertions; `analysis/verify_signed.py`, 536,988 assertions), and the elementary theory is additionally formalized in Lean. Statistical models were frozen on small scales and tested on uncomputed ranges. The single empirical claim on which the headline rests — the correlation $\mathrm{corr}(\ln f, \ln L(1,\chi_p)) = -0.62$ — was re-executed for this paper and is reported with its raw output in §8.2 and §10.

1.5 Plan of the paper

The argument is a single descent from the data to the law, and the reader may follow it as one thread. §3 establishes the empirical ground — the cross-validated solution counts and the lognormal law, confirmed by blind prediction. §4 supplies the structural reduction, the divisor-class kernel and the square obstruction (both formalized in Lean), and with it the exact reason the elementary toolbox stops. §5 is the pivot: stripped of its obvious trends, the residual count still carries a consistent quadratic-residue signal at every modulus. §6 and §7 then gather the law's two remaining ingredients and fix its limits — the **second moment**, which exposes the count as a two-shift divisor correlation and so locates the analytic wall (with one rigorous Erdős–Kac anchor), and the **growth exponent** 3, derived independently from the arithmetic geometry of the Cayley cubic. §8 assembles these into the L-function law, through a Fourier decomposition over Dirichlet characters and the McKee–Zhou singular series. §9 maps in full why the law stops short of a theorem; §10 records the three-layer source verification; §11 concludes. The main thread is §3 $\to$ §4 $\to$ §5 $\to$ §8; §6 and §7 run parallel to it and may be read after §8 without loss of continuity.

2. Notation and preliminaries

For a prime $p \geq 5$:

- $f(p)$ — number of solutions of $4/p = 1/x + 1/y + 1/z$ with $x \leq y \leq z$, $x, y, z \in \mathbb{Z}_{>0}$ (OEIS A192787).
- **Type I / Type II.** Every solution has least denominator $x \in (p/4, 3p/4]$. With $a = 4x - p$ and $B = px$, solutions are in bijection with divisors $d \mid B^2$, $d \equiv -B \pmod{a}$ (Lemma A, §4.1). Exactly two strata occur at primes: **Type I** ($p \nmid d$: $d \mid x^2$, $d \equiv -4x^2 \pmod{a}$) and **Type II** ($d = pd'$: $d' \mid x^2$, $d' \equiv -x \pmod{a}$). We write $f = 3f_I + 3f_{II}$ for the essentially-distinct counts of each stratum (the factor 3 is the placement symmetry of the p-divisible denominators).
- $\chi_p = (\frac{\cdot}{p})$ — the Legendre/Jacobi quadratic character; for $p \equiv 1 \pmod{4}$ it is the real character of conductor p attached to $\mathbb{Q}(\sqrt{p})$.
- $L(1, \chi_p) = \sum_{n \geq 1} \frac{\chi_p(n)}{n} = \prod_{\ell} \left(1 - \frac{\chi_p(\ell)}{\ell}\right)^{-1} = \frac{2\,h(p)\,\log \varepsilon_p}{\sqrt{p}}$ — the Dirichlet L-value, with $h(p)$ the class number and ε_p the fundamental unit of $\mathbb{Q}(\sqrt{p})$ (Dirichlet class-number formula).
- $\omega_3(n) = \#\{\ell \equiv 3\ (4) \text{ prime} : \ell \mid n\}$ — the **channel count**.

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3. The empirical landscape

3.1 Data and validation

We computed $f(p)$, with its Type I/II split, for all 278,570 primes of the six hard classes mod 840 from 73 to 2.01×10^9 — beyond the largest previously published per-prime dataset (3.5×10^7). Validation was exhaustive and is summarized below; no mismatch was found anywhere.

Check	Result
Rust vs C vs CUDA vs OpenMP engines, 5 scales	byte-identical
1-thread vs 16-thread; segmented vs single	byte-identical
Our engine vs the external published dataset (independent code/authors)	4,519 primes, 0 mismatches (incl. Type I/II)
Independent Python solution-enumerator vs both	exact on all targets

3.2 Zero-free, with a rising floor

Across all 278,570 primes, $f, f_I, f_{II} > 0$: the conjecture holds throughout, and **both mechanisms are always available.** The per-window minimum (the "floor") grows as

$$\min f \sim (\ln p)^{3.27\pm0.2}, \qquad \operatorname{median} f \sim (\ln p)^{3.30},$$

i.e. the floor and the median grow at the **same** exponent, itself consistent with the Elsholtz–Tao average order $\log^3 p$. The all-time record low is $f(2521) = 9$ ($2521 \equiv 1 \bmod 840$); no prime in the four following orders of magnitude approaches it. Meanwhile the relative dispersion $\sigma(\ln f)$ shrinks monotonically, $0.30 \to 0.144$ over five decades — the landscape *tightens* as it rises.

3.3 The lognormal law and four blind confirmations

Per dyadic window, $\ln f$ is normal to high precision, so window minima are pure order statistics of a lognormal with a rising mean and a shrinking variance. Fitting $\mu(\ln f)$ and σ on small scales, freezing them, and predicting the minima of then-uncomputed ranges gives a clean falsification protocol. Four predictions, each stated before its computation:

Range	Predicted	Observed
$[10^8, 2\times10^8]$	$\min f \in [175, 225]$	191
$[10^9, 1.01\times10^9]$	≈ 345	347
$[2\times10^9, 2.01\times10^9]$	median 681; $\min \in [351, 404]$	median 681 exact; $\min 405$
$[10^{10}, 10^{10}+10^7]$	$\min \in [439, 499]$; median ≈ 852	targeted min $= 534$ (safe side); $f(10000001041) = 980$, $f(10000003129) = 945$

The 10^{10} row required two new 128-bit engines to break the 2×10^9 counting wall; the measured floor 534 sits $\approx 7\%$ **above** the predicted band — the model under-predicts, i.e. the conjecture is even safer than the law says — and is consistent with the $\sim 10\%$ accuracy of the lognormal extreme-value model on held-out minima. The lognormal is the central empirical fact; everything that follows explains its mechanism.

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4. The kernel and the square obstruction (machine-verified)

The mechanism the lognormal demands begins not with analysis but with structure. Before any analytic input, $f(p)$ is an *exact* divisor count in prescribed residue classes — this is what makes it computable, and what will, in §§5–8, make it an L-value — while at the six square classes that count provably collapses in its coprime strata. Both facts are theorems, re-proved here in full and formalized in Lean 4 + Mathlib; they fix the floor of the landscape and the reason the elementary toolbox cannot raise it.

4.1 The divisor-class kernel (Lemma A)

> **Lemma A (kernel bijection).** For prime $p \geq 5$ and $x \in (p/4, 3p/4]$, put $a = 4x - p$, $B = px$ (so $\gcd(a, B) = 1$). Solutions of $4/p = 1/x + 1/y + 1/z$ with least denominator x are in bijection with divisors $d \mid B^2$, $d \leq B$, $d \equiv -B \pmod{a}$, via $y = (B + d)/a$, $z = (B + B^2/d)/a$.

The bijection is now machine-verified in Lean 4 + Mathlib **in both directions**, with no unproved steps. Forward (`esc_kernel`): from $a + p = 4x$, $B = px$, $de = B^2$ and $ay = B + d$, $az = B + e$, the identity $4/p = 1/x + 1/y + 1/z$ follows from the two clean steps $1/y + 1/z = a/B$ and $1/x + a/B = 4/p$. Converse (`esc_kernel_converse`): every solution yields $d = ay - B$, $e = az - B$ with $de = B^2$, because the equation *is* $a \cdot yz = B(y + z)$, whence $(ay - B)(az - B) = a^2yz - aB(y + z) + B^2 = B^2$. Forward and converse live in the **same** (x, d, a, B) coordinates as the obstruction below, so kernel and obstruction reason about one object.

4.2 Channels, identities, and their death at squares

A **channel** is a congruence rule producing a Type I or Type II solution for a whole residue class. Mordell's identities are precisely the class-wide channels; the Type II channel of "index m" exists for p iff the shifted integer $4pm + 1$ has a divisor $\equiv 3 \pmod{4}$ with the side-condition $m \mid y_1z_1$. The square classes are special because a prime $p \equiv \square \pmod{840}$ is a quadratic residue mod $3, 5, 7$, so **every channel whose congruence conditions live on the primes of** 840 **is dead**; channels on new primes fire for only $\sim 1/q$ of the class. This is Schinzel's theorem read channel-by-channel, and it is visible in data: floor primes' Type I spectra begin at $e = 116$ (for 2521) versus $e = 5$ for every covered-class prime.

4.3 The square obstruction (Lemma D), proved in full and in Lean

> **Lemma D.** Let $n = c^2$ (c odd), $x \in (n/4, 3n/4]$ with $\gcd(4x - n, nx) = 1$, $a = 4x - n \equiv 3 \pmod{4}$. Then no divisor of x^2 lies in the Type I class $d \equiv -4x^2 \pmod{a}$, and no divisor $d' \leq x$ of x^2 lies in the Type II class $d' \equiv -x \pmod{a}$: **the coprime strata are empty at odd squares.**

The proof rests on two Jacobi-symbol facts that pull in opposite directions:

1. **Required sign -1.** For each $\ell \mid a$, $\ell \nmid 2x$, so $\left(\frac{d}{\ell}\right) = \left(\frac{-4x^2}{\ell}\right) = \left(\frac{-1}{\ell}\right)$; multiplying over $\ell \mid a$ and using $a \equiv 3 \pmod{4}$ gives $\left(\frac{d}{a}\right) = \left(\frac{-1}{a}\right) = -1$. (Type II: $c^2 \equiv 4x\ (\ell)$ makes x a residue, so $\left(\frac{-x}{a}\right) = -1$ likewise.)
2. **Actual sign $+1$.** Any $d \mid x^2$ is $d = \delta^2 d_0$ with $d_0 \mid x$ squarefree, so $\left(\frac{d}{a}\right) = \left(\frac{d_0}{a}\right)$; reciprocity with $a \equiv 3 \pmod{4}$ and $a \equiv -c^2 \pmod{d_0}$ gives $\left(\frac{d_0}{a}\right) = +1$.

The contradiction empties the coprime strata. This is the coprime case of

Yamamoto's theorem — the case in which the engine and all covering arguments live.

Lean formalization. The file `erdos1/subsetsums/Subsetsums/ErdosStraus.lean` machine-verifies, with no unproved steps (every theorem depending only on Lean's standard axioms `propext, Classical.choice, Quot.sound`), 20 theorems spanning all three movements:

- *Sufficient conditions:* `esc_of_factorization`, `esc_of_K1` (Obláth's criterion: $4p+1$ has a divisor $\equiv 3 \bmod 4$), and the master `esc_of_typeII` ($(4y-1)(4z-1) = 4p\delta + 1$, $\delta \mid yz$).
- *The obstruction, all divisors and both strata:* `typeI_target_jacobi` and `typeII_target_jacobi` (the target classes force Jacobi symbol -1); `div_jacobi_one` (every $d \mid x^2$ coprime to a, **odd or even**, has Jacobi symbol $+1$); hence `typeI_obstruction` and `typeII_obstruction` derive `False`. The prime cores `four_sq_add_one_div_one_mod_four` and `eight_sq_add_one_div_one_or_three_mod_eight` discharge the K1/K2 channels by strong induction.
- *Chirality and positivity:* `typeI_neg_div_jacobi` (the obstruction is one-sided — a *negative* divisor is sign-compatible, so only the positive windows are emptied) and Theorem G (`esc_two_term_pos`, `esc_two_term_signed`): two unit fractions already give $4/p$, positive for $p \equiv 3$ (4) and signed for $p \equiv 1$ (4).

The development also records that the obstruction is **not** a disproof: $4/9 = 1/3 + 1/12 + 1/36$ holds at the square $n = 9$ even though its pure coprime strata are empty — solutions persist in mixed strata, exactly as Erdős–Straus requires.

4.4 No finite channel system (Theorem F) and the ε-covering cost

> **Theorem F.** If a channel valid on $n \equiv r \pmod m$ (with finitely many coprimality conditions, $n > n_0$) produces a coprime-stratum solution, then r is **not** a unit square $\bmod m$. Consequently any finite channel system with $M = \operatorname{lcm}$ of its moduli leaves every prime $p \equiv 1 \pmod M$ uncovered — a set of relative density $1/\varphi(M) > 0$.

Proof. If $r \equiv u^2 \pmod m$, CRT produces odd $c \equiv u$ with $n = c^2$ in the channel's class, where Lemma D forbids a coprime-stratum solution. The six square classes mod 840 are the $M = 840$ cross-section. ∎

A complementary ε-covering theorem (Theorem E) shows that explicit channel families can cover all but a density-ε set of primes, but at cost $T \approx \exp(\varepsilon^{-2})$ identities (the uncovered density falls only as $(\log T)^{-1/2}$, by Mertens). **The square classes can never be closed by identities, covering congruences, or the circle method** — a structural reason, not a failure of ingenuity, for eighty years of plateau. What a proof needs instead is a *pointwise lower bound for divisors of shifted integers in prescribed residue classes, uniform in p* — available today only on average and on density-1 sets.

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5. The residual spectrum: a quadratic-character fingerprint

This is the pivot from "the elementary toolbox fails" to "an L-function governs the count." After removing the window trend and the mod 840 class means from $z = \ln f$, the residual is not noise: the residue $p \bmod q$ still predicts z at **every** prime modulus $q \le 199$ coprime to 840, and always with the same sign. Define the quadratic-residue contrast

$$s_q = \text{mean } z(p \text{ non-residue } \bmod q) - \text{mean } z(p \text{ residue } \bmod q).$$

Then $s_q > 0$ at 19 of 21 moduli $q \le 97$ (the two flat ones have the smallest predicted effects), with a measured decay law

$$s_q \;\approx\; 18.2\, q^{-1.95}.$$

"Non-residues richer at every modulus" is exactly the statement that $f(p)$ carries the multiplicative signal of the Legendre symbol $\left(\frac{p}{q}\right)$ at every prime q, with a *consistent sign* — being a residue $\bmod\, q$ prunes channels (Theorem F seen from below). An additive model over these contrasts saturates out-of-sample at $\approx 58\%$ of the residual variance, the remaining $\approx 42\%$ being carried by the prime factorizations of the shifted integers $4pm+1$ themselves. The q^{-2}-summable ladder over a consistent quadratic signal is the empirical silhouette of an Euler product $\prod_q \left(1 - c\left(\frac{p}{q}\right)/q\right)$ with a scalar $c > 0$; §8 confirms it *is* one.

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6. The second moment and the analytic core

§5 located the signal; turning it into an L-value (§8) needs two further inputs, supplied by this section and the next. Here we take the analytic route, through the **second moment** of f. It pays twice. It exposes the precise arithmetic object the count is built from — a correlation of two shifted divisor sums — and, in the same breath, locates the wall that stops a proof. The section then closes on firmer ground, with the one fully rigorous distributional law within reach: an Erdős–Kac theorem that anchors the §3 lognormal in the second moment and identifies the floor primes exactly.

6.1 The two-shift Titchmarsh reduction (exact)

A Type II solution corresponds bijectively to (y', z', δ) with

$$(4y'-1)(4z'-1) = 4p\delta + 1, \qquad \delta \mid y'z', \qquad p = \frac{(4y'-1)(4z'-1)-1}{4\delta},$$

so $f_{II}(p)$ counts divisors $\equiv 3 \pmod 4$ of the shifted integers $\{4p\delta+1\}_\delta$ with the divisibility side-condition (machine-verified on every Type II solution of 499 primes). Writing $r_\delta(p)$ for the count at fixed δ, unfolding the square gives the exact identity

$$\sum_{p\le N} f_{II}(p)^2 = \sum_{\delta_1,\delta_2} \sum_{\substack{p\le N\\ p \text{ prime}}} r_{\delta_1}(p)\, r_{\delta_2}(p),$$

a correlation of two divisor-type functions of the linear forms $4\delta_1 p+1$, $4\delta_2 p+1$ over primes — a **two-shift Titchmarsh divisor problem**. This is the explicit form of Elsholtz–Tao's Remark 1.3, apparently written down here for the first time.

6.2 The wall hardens with scale

Measured on exact δ-decompositions, the sum is 98.5% **off-diagonal** ($\delta_1 \neq \delta_2$), and the off-diagonal share *rises* with p: $0.9554\,(10^4) \to 0.9754\,(10^5) \to 0.9836\,(10^6)$, because distinct Type II solutions occupy near-unique shifts (δ-multiplicity ≈ 1.08, constant). With $f_{II} \asymp (\log p)^3$ the diagonal fraction falls like $C/(\log p)^3 \to 0$: the second moment is *asymptotically pure two-shift Titchmarsh*. The required level of distribution — conductor $\sim p^2$, two simultaneous shifts, over primes — sits past Bombieri–Vinogradov ($p^{1/2}$), past the ternary-divisor records (Sharma $p^{1/2+1/30}$ for a single modulus; Aydemir–Boran $p^{8/11}$ averaged), and past the GRH-conditional Titchmarsh power-savings (Drappeau; Assing–Blomer–Li). A pointwise version is additionally **parity-blocked** — the classical Selberg/$\sqrt{x}$ parity barrier, which any unconditional handling of the divisor function over primes must break via a Type-II/bilinear input (the line of Granville–Shao, *beyond the $x^{1/2}$-barrier for multiplicative functions in APs*, arXiv:1703.06865). This is the analytic mirror of Theorem F.

6.3 The rigorous anchor: an Erdős–Kac channel law

The one rigorous distributional statement available is a theorem, not a heuristic. Since $\sum_{\ell\equiv 3(4)} 1/\ell \sim \frac{1}{2}\ln\ln x$ and the prime factors of $4p+1$ equidistribute $\bmod 4$ (Bombieri–Vinogradov), the Kubilius–Shapiro / Halberstam (1956) central limit theorem for additive functions of shifted primes gives:

$$\omega_3(4p+1) \text{ is asymptotically Normal with mean } \sim \text{ Var } \sim \tfrac{1}{2}\ln\ln p.$$

Measured: mean 1.056, Var 1.072 (the Erdős–Kac signature), skew $\to 0$. The K1 channel fires iff $\omega_3(4p+1) \geq 1$, so **K1-starvation** $\iff \omega_3(4p+1) = 0$ — a Landau–Ramanujan event of density $\sim C/\sqrt{\ln p}$ (measured 0.473), and all four documented floor primes $(2521, 4201, 9601, 20521)$ have $\omega_3(4p+1) = 0$. The floor is the **left-tail large deviation of an Erdős–Kac variable**. Crucially this also settles the distributional shape: Var/mean grows $3.5 \to 13.8$ (so f

is over-dispersed, *not* Poisson), while $\mathrm{Var}/\mathrm{mean}^2 \approx e^{\sigma^2} - 1$ holds to $\approx 5\%$ in the largest windows — f is **lognormal**, confirming §3.3 in the second moment. Why it does not prove ESC: an Erdős–Kac law gives a left tail that is *thin* (density $\to 0$) but provably *non-empty*, and "thin but non-empty" is exactly the density-1$\to$all-primes gap.

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7. The growth exponent from the Cayley cubic (heuristic)

We return to the wall in §9; the law's one remaining ingredient is its exponent. The 3 in $f(p) \sim (\log p)^3$ — so far an empirical fact (§3) matching the Elsholtz–Tao average order — has an independent arithmetic-geometry derivation. The Erdős–Straus surface $4xyz = n(xy + yz + zx)$ has open part $V_n \cong \mathbb{G}_m^2$ (Bright–Loughran, 2020: geometric unit group of rank 2), with a boundary triangle of lines at infinity. Applying the Browning–Wilsch (2025) Batyrev–Manin–Peyre heuristic for integral points on log-K3 cubic surfaces,

$$N_U^\circ(B) \;\sim\; c\,(\log B)^{\varrho_U + b},$$

with $\varrho_U = \operatorname{rank}\operatorname{Pic}(U)$ and b the maximal number of boundary components meeting at a real point, and calibrating against the authors' own Markoff computation (triangle boundary, exponent 2 = Zagier's law), the Cayley cubic's rank-2 unit group gives $\varrho_U + b = 3$. (Browning–Wilsch treat the Markoff and sum-of-three-cubes surfaces; the calibration to the Cayley/Erdős–Straus cubic is ours, not a result of theirs.) With height $B \asymp p$,

$$f(p) = N_{U_p}^\circ(B) \;\sim\; c\,(\log p)^3,$$

matching the Elsholtz–Tao average and, on the repo's own data, $k = 3.03$ for $\ln(\operatorname{median} f) = k \ln\ln p + b$ over six decades. Two honest hedges: b is model-dependent (pinning it rigorously means a Tamagawa/Clemens computation on the weak-dP3 model neither paper has done), and the framework is provably **blind to the square classes** — a direct count gives the cubic exactly $\ell^2 + 1$ points mod ℓ independent of n's quadratic-residue status. The geometry fixes the exponent and the typical size; it sees nothing of existence. The square-class suppression is the finer character/parity effect that §8 identifies.

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8. The L-function law

Every piece is now in hand: the quadratic-residue ladder of §5, the second moment built from divisor sums of quadratics (§6), and the exponent 3 (§7). This section assembles them into the law of the abstract. The ladder sharpens into a single Fourier coefficient (§8.1); that coefficient's Euler product is, by definition, $L(1,\chi_p)$ (§8.2); and McKee–Zhou supplies the rigorous precedent that divisor sums of quadratics are governed by exactly this L-value (§8.3). The destination promised at the end of §5 is reached here.

8.1 The character decomposition

The local divisor density of $f(p)$ at a prime $\ell \nmid 840$ — measurable because the hard primes equidistribute $\bmod \ell$ — was Fourier-decomposed over the Dirichlet characters $\bmod \ell$. At **every** $\ell \in \{11, 13, 17, 19, 23\}$ the dominant non-trivial character is the **quadratic (Legendre) character** (p/ℓ), with a *negative* coefficient of order $1/\ell$ (its size is the scale-dependent exponent c of §8.2), at 14–47σ (166,000 primes; `analysis/lfunction_connection.py`, `class_local_density.py`, `sigma7_char_fit.py`). The mechanism is transparent: $f(p)$ counts divisors of the shifted integers $4p\delta+1$ in residue classes, and the local density of such divisors at ℓ is governed by how ℓ splits in $\mathbb{Q}(\sqrt{p})$, i.e. by $\chi_p(\ell) = (p/\ell)$. The §5 ladder $s_q \approx 18\,q^{-1.95}$ is exactly the first-order ($O(1/\ell)$) shadow of this signal.

8.2 The law

The product of these quadratic characters is, by definition, the Euler product of the quadratic L-function:

$$f(p) \;\approx\; (\log p)^3 \cdot \prod_{\ell} \Big(1 - \frac{c\,\chi_p(\ell)}{\ell}\Big) \;\approx\; (\log p)^3 \cdot L(1,\chi_p)^{-c}, \qquad c > 0.$$

Here $\chi_p(\ell) = (\frac{p}{\ell})$ is the Legendre symbol of §2 and $c > 0$ is a single scalar exponent — **not** a function of ℓ (its *fitted value* drifts with scale; Table 1). The two displayed forms agree to first order in $1/\ell$: the Euler product gives $\prod_\ell \big(1-\chi_p(\ell)/\ell\big) = L(1,\chi_p)^{-1}$ exactly, and $(1-x)^c = 1 - cx + O(x^2)$, whence $\prod_\ell \big(1 - c\,\chi_p(\ell)/\ell\big) = L(1,\chi_p)^{-c}\,(1+O(1))$ — the convergent $O(1/\ell^2)$ remainder is absorbed into the amplitude. This is an **empirical law** ($f(p)$ carrying the character signal with mean order $(\log p)^3$), not a claimed identity; the $(\log p)^3$ is the Elsholtz–Tao average order (and §7's Cayley-cubic count), independent of the L-modulation it multiplies. By Dirichlet's class-number formula $L(1,\chi_p) = 2h(p)\log\varepsilon_p/\sqrt{p}$, this says $f(p)$ **is modulated by the class number / regulator of $\mathbb{Q}(\sqrt{p})$: larger $L(1,\chi_p) \Rightarrow$ fewer Erdős–Straus solutions.** A direct test against an independent truncated $\ln L(1,\chi_p) = -\sum_{\ell\le X}\log\big(1-(p/\ell)/\ell\big)$ — built with no reference to the f-data — gives, on 14,955 hard primes near 10^9 (re-executed for this paper; output of `analysis/lfunction_connection.py`, where `#Euler factors` is the number of primes $\ell \le X$ in the truncated product, **not** the sample size):

```
n = 14,955 hard primes near 1e9   (p in [1.000, 1.010]e9, ln p ~ 20.73,
     so the (log p)^3 trend is ~constant across the slice)
```

```
truncation X   #Euler factors (l<=X)   corr(ln f, ln L)   95% CI (Fisher z)
      50              11                  -0.5977          [-0.6079, -0.5873]
     200              42                  -0.6173          [-0.6271, -0.6073]
     500              91                  -0.6200          [-0.6298, -0.6100]
    1500             235                  -0.6199          [-0.6296, -0.6099]

partial corr(ln f, ln L | ln p) = -0.6199   ((log p)^3 trend removed)
regression slope d(ln f)/d(ln L) = -0.527   (an estimate of -c)
```

The correlation is -0.62 (95% CI $\approx \pm 0.01$ at $n = 14{,}955$), stable across truncation and — because the slice spans only $\ln p \in [20.723, 20.733]$ — **unchanged when the** $(\log p)^3$ **trend is partialled out** (partial $r = -0.62$): it is the L-value, not the size of p, that f tracks. But this -0.62 is the value **at** 10^9, and the exponent it implies is not universal. Measured in narrow windows from 10^6 to 10^{10} (Table 1 below; the 10^{11} row awaits a multi-day GPU run still in progress), the correlation itself is nearly scale-stable — it stays within a ± 0.03 band of -0.62, with a shallow (but statistically resolved) maximum near $p \sim 10^7$–10^8 and only a gentle weakening beyond it (-0.620 at 10^9, -0.596 at 10^{10}). What moves with scale is the *dynamic range* of f: $\sigma(\ln f)$ shrinks monotonically (from 0.21 down to 0.14) as the count distribution tightens, while $\sigma(\ln L)$ climbs from ≈ 0.13 and saturates near 0.175 by 10^8. The implied $c = |\mathrm{corr}| \cdot \sigma(\ln f)/\sigma(\ln L)$ therefore **declines monotonically** across the whole range — from 0.93 at 10^6 to 0.46 at 10^{10} — driven by the narrowing spread of f, not by any collapse of the L-coupling (whose correlation barely moves). Only the **sign** is scale-robust — negative at every scale tested; the magnitude is a finite-size quantity, and whether c tends to a positive limit or to 0 as $p \to \infty$ we cannot yet decide, though the monotone descent over four decades is, if anything, weak evidence for the latter. The six hard square classes mod 840 are a *finer* shadow of the same object: at $\ell = 5, 7$ the prime p is forced to be a residue, so the leading quadratic character is constant and the class-splitting is carried by the higher residue characters — a cubic character at 7 plus a $\sim 17^\circ$ **chiral phase** ($\sigma_7(c) = a_0 + 2\,\mathrm{Re}(b\,\psi(c))$, $b = 8.4\,e^{163^\circ i}$), the local shadow of the signed-sector see-saw and impossible for any single Dirichlet character.

Table 1. The L-correlation and the implied exponent across scale. Narrow windows (so $\log p$ is held fixed within each), computed cleanly from scratch with a single validated engine (`fp128` mode 2; driver `analysis/run_scale_study.sh`, table by `analysis/scale_table.py`). Here $f = f_{\mathrm{I}} + f_{\mathrm{II}}$ and $\ln L$ is the truncated quadratic L-value at $X = 1500$ (235 Euler factors); the 10^9 row reproduces the headline -0.620. The correlation is nearly scale-stable (shallow maximum near 10^7–10^8), $\sigma(\ln f)$ shrinks monotonically and $\sigma(\ln L)$ saturates near 0.175 by 10^8, so the implied $c = |\mathrm{corr}|\,\sigma(\ln f)/\sigma(\ln L)$ **declines monotonically** with scale. The sign is negative at every scale; the 95% CI on each correlation is $\approx \pm 0.01$ at $n \approx 14{,}000$ and $\approx \pm 0.03$ at $n \approx 2{,}000$.

scale p	n	$\sigma(\ln f)$	$\sigma(\ln L)$	$\mathrm{corr}(\ln f, \ln L)$	implied c
10^6	1,927	0.208	0.132	−0.588	0.925
10^7	2,161	0.185	0.161	−0.646	0.741
10^8	2,001	0.166	0.170	−0.640	0.625
10^9	14,955	0.148	0.174	−0.620	0.527
10^{10}	13,564	0.135	0.175	−0.596	0.458
10^{11}	—	—	—	—	—

(Rows 10^6–10^{10} are the completed fresh recomputation; the 10^{11} row awaits a multi-day GPU run still in progress and is filled on completion.)

8.3 The mechanism: McKee–Zhou and the Gauss–Siegel precedent

The rigorous engine behind the law is classical. For an irreducible quadratic F, the divisor sum $\sum_{n\le N} \tau(F(n))$ has singular series

$$\mathfrak{S}(F) = \frac{2\,L(1, \chi_{\mathrm{disc}\,F})}{\zeta(2)}$$

(McKee, *Math. Proc. Camb. Phil. Soc.* **126** (1999); explicit constant in Zhou, arXiv:1611.10186; cf. Lapkova). This is the divisor-count analogue of the Gauss–Siegel fact that ternary representation counts are governed by quadratic L-values ($r_3(n) \propto H(n) \propto L(1, \chi_{-n})$). Elsholtz–Tao construct $f(p)$ from precisely such divisor sums $\tau(kab^2+1)$ but use McKee only as an O-bound, never extracting the constant — which is exactly where $L(1, \chi_p)$ lives. Because an $L(1, \chi_p)^{-c}$ factor has mean $\approx$ const, the law leaves the Elsholtz–Tao first moment (of order $N \log^2 N$) untouched while explaining the **prime-by-prime variance** a first-moment bound is blind to.

8.4 What the L-lens buys

Three consequences are immediate and structurally important, granting the law:

1. **The exceptional set is theorem-controlled.** Thin $f \iff$ large $L(1, \chi_p)$. By Granville–Soundararajan, $\#\{p \le N : L(1, \chi_p) \ge e^\gamma \tau\}$ falls **super-exponentially** in τ. Any Erdős–Straus exception must lie in the **extreme class-number tail** — a provably super-rare, density-controlled set. (Compare the unconditional "almost all" of Vaughan, 1970.)
2. **Siegel zeros are the best case, not the worst.** A Siegel zero $\beta \to 1$ forces $L(1, \chi_p)$ *small*, hence $f(p)$ *large*. The conjecture's adversary is the opposite extreme — maximal L, of order $e^\gamma \log\log p$ — and even there the budget gives $f \gg (\log p)^3/(\log\log p)^c \to \infty$. **ESC is robust to the classic Landau–Siegel nightmare.**
3. **The size budget never threatens $f > 0$.** The best unconditional ceiling $L(1, \chi_p) \le (0.197 + o(1)) \log p$ (Stephens) gives, granting the law,

$f \gg (\log p)^{3-c} \to \infty$ (the measured $c < 1$ makes the exponent exceed 2; and were $c \to 0$ with scale, the bound would only strengthen toward $(\log p)^3$). The entire difficulty is upgrading $\asymp$ to a proven inequality — never the magnitudes.

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9. Honest status: a verified law, short of a theorem

The payoffs of §8.4 are real but conditional on the law. This section discharges that condition as far as it will go, and marks precisely where it will not: a derivation of the singular series blocks at the δ-split (§9.1), the asymptotic and the matching lower bound are parity-obstructed (§9.2), and only a low-payoff upper bound survives parity-safe (§9.3). The wall met here is the same one §6.2 measured, now read from the side of a proof.

9.1 Where a derivation blocks (McKee–Zhou and the δ-split)

$L(1, \chi_p)$ here is the **singular series itself**, not an external special value: bounding it controls the *main term*, which it already predicts. The unproven content is the *error term* — that the two-shift divisor sum is asymptotic to its singular series with a power saving, at conductor $\sim p^2$, over primes. Attempting to *derive* the singular series via McKee–Zhou meets a verified obstruction. McKee–Zhou needs the count to be $\sum \tau(F(n))$ for a **single** irreducible quadratic F; but the Type II count carries the side-condition $\delta \mid y'z'$, and δ splits *arbitrarily* across y' and z' in real solutions — at $p = 2521$ the solution with $\delta = 98 = 2 \cdot 7^2$ has $\gcd(\delta, y') = \gcd(\delta, z') = 14$, dividing neither factor alone. A direct reduction to one quadratic divisor sum therefore **undercounts** (the clean conic form returns $f_{II}(2521) = 1$ versus the true 3). The split is exactly what makes f_{II} a **two-shift divisor correlation**, not a single $\sum \tau(F)$: the L-value does not factor out through one class-number formula. The effective exponent c (scale-dependent — §8.2 — and negative-signed, i.e. Yamamoto suppression) is the *aggregate* character content of that correlation.

9.2 The parity wall, and why upper bounds are the only safe target

By Cauchy–Schwarz, $\#\{p \leq N : f(p) > 0\} \geq (\sum_p f)^2 / \sum_p f^2$. The numerator is the Elsholtz–Tao first moment (order $N \log^2 N$); the denominator is the §6 second moment, of expected order $N \log^5 N$. A one-sided **upper** bound $\sum_p f^2 \ll N(\log N)^5$ is the rare parity-*safe* target — Selberg's barrier blocks asymptotics and lower bounds, not majorants — and is winnable in principle by divisor-correlation technology (Nair–Tenenbaum; Henriot) applied to $d(4\delta p + 1)$, with a Brun–Titchmarsh majorant on the single prime constraint. We verified (§18.7 of the report) that the resulting δ-split correlation upper bound is genuinely parity-safe and structurally winnable. But its only payoff is Cauchy–Schwarz — a **positive proportion** $1/C$ with $C = \mathbb{E}[f^2]/\mathbb{E}[f]^2$ — which is *weaker* than the **density 1** already given unconditionally by Vaughan (1970). And it delivers only the *order* of the second moment, not the leading constant, so it does not even establish the $L(1, \chi_p)$ law rigorously.

9.3 The complete map

The valuable statements — the asymptotic second moment, the exact $L(1, \chi_p)^{-c}$ singular series, the L-controlled exceptional set — all require the matching **lower** bound $f \gg (\log p)^3 L^{-C}$, which is the two-shift Titchmarsh asymptotic, which is parity-blocked. So the honest map is:

Target	Parity	Winnable?	Payoff
Upper bound $\sum_p f^2 \ll N \log^5 N$	safe	yes (research-level)	positive proportion ($\leq$ Vaughan), order only
Asymptotic second moment	blocked	no	density 1, the L-constant
Lower bound $f \gg (\log p)^3 L^{-C}$	blocked	no	the L-law as a theorem; with $L \leq \frac{1}{2} \log p$, ESC for the class
Pointwise / nonempty-left-tail	blocked	no	the conjecture

One parity-safe but low-payoff target, and a fence of parity-blocked high-value ones beyond it. **The session's genuine advance is the $L(1, \chi_p)$ lens itself** — it proves the magnitudes are never the obstacle, confines any counterexample to a super-rare L-extreme set, and isolates the single needed inequality cleanly — *not* a new bound.

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10. Provenance and source verification

Because the paper's value depends on building correctly on prior work, every cited source was verified at **three layers**: (i) first-party confirmation that each arXiv identifier resolves to a real paper with the stated authors, title, and date (we fetched the abstract pages directly — every identifier below was confirmed); (ii) first-party theorem-level confirmation that the specific result we lean on is genuinely the paper's (we checked the load-bearing claims against primary sources and explicit statements); (iii) an independent four-agent web audit (arXiv API, ar5iv full text, journal DOIs) run in parallel. That audit has **completed and agrees**: **every cited source resolves to a real, on-topic paper, correctly attributed — no fabricated or hallucinated citation exists** (including all five post-2025 identifiers). The central empirical claim was additionally **re-executed** (the raw table in §8.2). The corrections below — all already folded into the text above — are matters of exact title, volume, dating, and author-name order, plus two claim-precision fixes (the Elsholtz–Tao first moment, and crediting the McKee constant to McKee rather than to Elsholtz–Tao). **None touches the substance of the L-function law or the wall.**

Corrections folded in (from the verification pass):

- **arXiv:2509.00128** is *Further verification and empirical evidence for the Erdős–Straus conjecture* by **Spiridon Mihnea and Dumitru C. Bogdan** (29 Aug 2025), verifying to $\mathbf{10^{18}}$ and tabulating $f(p)$ — resolving a three-way author/ceiling confusion (10^{17} was Salez's separate result).
- **Granville–Shao** (arXiv:1703.06865) is **2017** (rev. 2019), *not* 2023; its explicit content is multiplicative functions in APs *beyond the $x^{1/2}$-barrier* (level $x^{20/39}$). The "parity" point we attach to it is the **classical $\sqrt{x}$/Selberg parity barrier** that this line of work engages, stated as such, not a verbatim "every-modulus equidistribution is false" theorem.
- Exact titles (the report sometimes paraphrased): Zhou 1611.10186 = *The explicit asymptotic formula of divisor function on average over values of quadratic polynomial*; Sharma 2303.06087 = *Bilinear sums with $GL(2)$ coefficients and the exponent of distribution of d_3* ($1/2 + 1/30$ confirmed); Drappeau 1504.05549 = *Sums of Kloosterman sums in arithmetic progressions, and the error term in the dispersion method*; Aydemir–Boran 2601.12601 = *Improved Averaged Distribution of $d_3(n)$ in Prime Arithmetic Progressions* (8/11 confirmed); Topacoğullari 1512.05770 = *On a certain additive divisor problem* (Acta Arith. 181, 2017).
- **McKee** (1999, MPCPS **126**, 17–22): his $\lambda = 12H^*(\Delta)\log\varepsilon_\Delta/(\pi^2\sqrt{\Delta})$ for $\Delta > 0$ equals $2L(1,\chi_\Delta)/\zeta(2)$ by the class-number formula — the mechanism of §8.3 is exact. Explicit version: **Lapkova**, arXiv:1704.02498.
- **Granville–Soundararajan**, *The distribution of values of $L(1,\chi_d)$*, GAFA **13** (2003) 992–1028: the large-value proportion decays **double-exponentially** in τ — stronger than the "super-exponential" we claim.
- **Stephens** (*Optimizing the size of $L(1,\chi)$*, Proc. LMS (3) **24** (1972) 1–14): $|L(1,\chi)| \le \frac{1}{2}(1 - e^{-1/2} + o(1))\log q$, and $\frac{1}{2}(1 - e^{-1/2}) = 0.1967\ldots$ — the constant 0.197 is exact, prime conductor (extended to all moduli by Pintz; recent explicit $\frac{1}{2}\log q$).
- **Elsholtz–Tao first moment** is the two-sided bound $N\log^2 N \ll \sum_{p\le N} f(p) \ll N\log^2 N\log\log N$ — **not** a clean $\asymp N\log^2 N$; the average order is $\log^3 p$ only up to the $\log\log$ factor they note but cannot remove. Corrected wherever stated.
- **The McKee singular-series constant is credited to McKee, not to Elsholtz–Tao.** ET's contribution is the divisor-sum realization of $f(p)$; the constant $2L(1,\chi_{\mathrm{disc}})/\zeta(2)$ is McKee's (via the class-number formula). We no longer phrase ET as "invoking McKee."

- **Halberstam** — the Erdős–Kac CLT for additive functions of *shifted primes* is Part **III**, *J. London Math. Soc.* **31** (1956) 14–27 (Part I is **30** (1955) 43–53). Corrected from "30 (1955)."
- **Bright–Loughran** — "no Brauer–Manin obstruction" holds *for the existence of solutions* (Thm 1.1); their Brauer group is the nontrivial $\mathbb{Z}/2\mathbb{Z}$ (Thm 1.6) and does obstruct *strong approximation* (Thm 1.8). Stated with that qualifier; the existence claim is what makes the difficulty analytic.
- **Browning–Wilsch** treat the Markoff and sum-of-three-cubes surfaces; the calibration to the Cayley/Erdős–Straus cubic giving exponent 3 is **our** application of their heuristic, presented as such, never as their stated result.

Source	We use it for	Verification
Elsholtz–Tao, *J. Aust. Math. Soc.* 94 (2013), arXiv:1107.1010	$f(p)$; first moment $N\log^2 N \ll \Sigma \ll N\log^2 N \log\log N$; Type I/II split; Remarks 1.2–1.3; density-1 lower bound (Thm 1.8)	✓ ID + claim (was $\asymp$ -> two-sided)
Vaughan, *Mathematatika* 17 (1970)	almost-all exceptional set $\ll N\exp(-c(\log N)^{2/3})$	✓ standard
Mordell, *Diophantine Equations* (1969)	the six square classes mod 840; covering identities	✓ standard
Schinzel (2000); Yamamoto	no identity family covers a class with squares; $f_I = f_{II} = 0$ on odd squares	✓ standard (+ Lean re-proof)
Salez (2014), arXiv:1406.6307	seven modular equations; verification to 10^{17}	✓ ID + claim
McKee (1999, MPCPS 126); Zhou 1611.10186; Lapkova 1704.02498	singular series $\sum \tau(\text{quad.}) = 2L(1,\chi_{\text{disc}})/\zeta(2)$	✓ ID + theorem
Gauss; Siegel	$r_3(n) \propto H(n) \propto L(1,\chi_{-n})$ (the L-value precedent)	✓ classical
Granville–Soundararajan, GAFA 13 (2003)	double-exponential tail of $L(1,\chi)$ values	✓ ID + theorem
Stephens (+ Pintz)	$L(1,\chi_p) \leq (0.197+o(1))\log p = \frac{1}{2}(1-e^{-1/2})\log p$	✓ constant exact
Granville–Shao, arXiv:1703.06865 (2017/19)	level beyond $x^{1/2}$ for multiplicative functions; the $\sqrt{x}$ parity barrier	✓ ID; parity = classical barrier

Source	We use it for	Verification
Halberstam, *JLMS* **31** (1956) 14–27 (Part III)	Erdős–Kac CLT for additive functions of shifted primes	✓ (vol/yr corrected)
Ford, *Annals* 168 (2008), arXiv:math/0401223	lognormal divisor-in-interval law	✓ ID + claim
Browning–Wilsch, *Selecta Math.* 31 (2025), arXiv:2407.16315	$(\log B)^{\varrho_U+b}$ integral-point heuristic; Markoff calibration	✓ ID; Cayley/exponent-3 is **our** application
Bright–Loughran, *Bull. LMS* 52 (2020), arXiv:1908.02526	$V_n \cong \mathbb{G}_m^2$; no BM obstruction *to existence* ($\mathrm{Br} = \mathbb{Z}/2\mathbb{Z}$ obstructs strong approx.)	✓ ID + claim + qualifier
Sharma 2303.06087; Aydemir–Boran 2601.12601; Drappeau 1504.05549; Assing–Blomer–Li 2005.13915; Topacoğullari 1512.05770	divisor-in-AP / Titchmarsh records that fall short of the two-shift need	✓ all IDs + levels
Mihnea–Bogdan (2025), arXiv:2509.00128	external $f(p)$ dataset (cross-validated); 10^{18} frontier	✓ ID + authors fixed
Ventas 2605.04551; Bello-Hernández–Benito–Fernández 2606.10922; Mballa 2602.20036	CF reformulation; divisor parametrization; density-one parametrization (all heuristic, hit parity)	✓ all IDs + content
Pomerance–Weingartner (2025), arXiv:2511.16817	exceptions to the Erdős–Straus–Schinzel conjecture	✓ ID + claim

The faithful-use principle applied throughout: where a cited result is used **rigorously** (McKee–Zhou's singular series, Halberstam's CLT, Bright–Loughran's geometry, Vaughan's exceptional set, Granville–Soundararajan's and Stephens' L-value bounds, the Lean-internal lemmas), it is invoked as a theorem; where it is used **heuristically or by extrapolation** (Browning–Wilsch's exponent, the lognormal extreme-value model, the L^{-c} correlation as a law), it is labeled as such and never upgraded. No source is cited for a stronger statement than it proves.

11. Conclusion

The Erdős–Straus solution count $f(p)$ is governed, to the accuracy we can measure and reproduce, by the quadratic L-value $L(1, \chi_p)$ — the class number and regulator of $\mathbb{Q}(\sqrt{p})$. We reached this not by guessing but by a documented descent: the lognormal law and its blind confirmations established that f has a clean distribution; the machine-verified kernel and square obstruction established that f is a divisor count in residue classes and pinned why elementary methods stop; the residual spectrum exposed a consistent quadratic-residue ladder at every modulus; and the character decomposition turned that ladder into an Euler product, hence into $L(1, \chi_p)^{-c}$, with the McKee–Zhou singular series as the rigorous precedent.

The law is verified and structurally identified, but it is not a theorem, and we have mapped exactly why: the underlying object is a two-shift Titchmarsh divisor correlation whose asymptotic and lower-bound content is parity-obstructed, leaving only a low-payoff upper bound as parity-safe. What the law nonetheless settles, conditionally on itself and consistent with everything unconditional, is the *shape* of the problem: the magnitudes never threaten $f > 0$, Siegel zeros help rather than hurt, and any counterexample is confined to the extreme class-number tail — a super-rare, density-controlled set. That is a sharper picture of where an Erdős–Straus exception could possibly hide than the conjecture has had, and it points the next rigorous effort at a single, cleanly isolated inequality.

Appendix A. The Lean 4 + Mathlib development

`erdos1/subsetsums/Subsetsums/ErdosStraus.lean` formalizes the elementary theory completely, with no unproved steps; `#print axioms` reports only Lean's standard `propext, Classical.choice, Quot.sound` for every theorem. The 20 theorems, by role:

- **Kernel:** `esc_kernel`, `esc_kernel_converse` (Lemma A, both directions).
- **Sufficient conditions:** `esc_of_factorization`, `esc_of_K1` (Obláth), `esc_of_typeII` (master Type II), `comp_mod_four` (helper).
- **Obstruction (prime cores):** `prime_factor_four_sq_add_one`, `four_sq_add_one_div_one_mod_four`; `prime_factor_eight_sq_add_one`, `eight_sq_add_one_div_one_or_three_mod_eight`.
- **Obstruction (general Lemma D):** `typeI_target_jacobi`, `typeII_target_jacobi` (target classes force -1); `typeI_div_jacobi_one`, `div_jacobi_one` (every divisor of x^2 gives $+1$, no parity restriction); `typeI_obstruction`, `typeII_obstruction` (the contradictions).
- **Chirality and positivity:** `typeI_neg_div_jacobi` (one-sidedness); `esc_two_term_pos`, `esc_two_term_signed` (Theorem G).

Non-vacuity is witnessed by worked examples (4/2 via K1; the obstruction biting at $n = 9$; $4/9 = 1/3 + 1/12 + 1/36$ persisting in mixed strata). Build/verify:

`cd erdos1/subsetsums && lake exe cache get && lake build`. Numeric cross-checks: `analysis/verify_lemmas.py` (8,719 assertions).

Appendix B. Reproducing the headline

```
# the L(1,chi_p) correlation of §8.2 (needs sympy; reads data/hard_1e9_slice.csv)
python3 analysis/lfunction_connection.py
#   -> corr(ln f, ln L) = -0.60 ... -0.62 across X = 50 ... 1500; slope -0.527 (at p~1e9;
#      the implied exponent c is scale-dependent - see the §8.2 scale table)

# the character decomposition (dominant quadratic (p/l) at l = 11..23, 14-47 sigma)
python3 analysis/class_local_density.py
python3 analysis/sigma7_char_fit.py

# the second-moment reduction and Erdős-Kac channel law (§6)
python3 analysis/second_moment.py
python3 analysis/offdiag_scaling.py
```

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Full derivations, tables, and machine checks: `REPORT.md` §§2–18; figures in `plots/`.